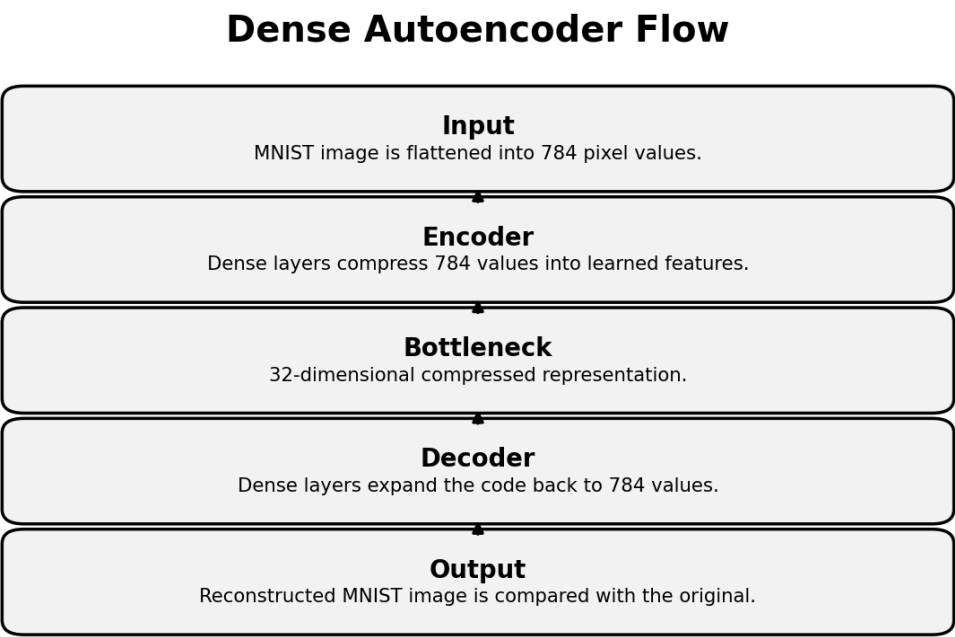


Learning Day 6 Autoencoders

Variable, Method, and Tensor Flow for the Three Completed Notebooks

This document explains how data moved through the three completed notebooks. It is designed to answer: where does each variable come from, what method or layer is applied, what output is created, what shape changes happen, and why the step matters.

1. Dense Autoencoder on MNIST



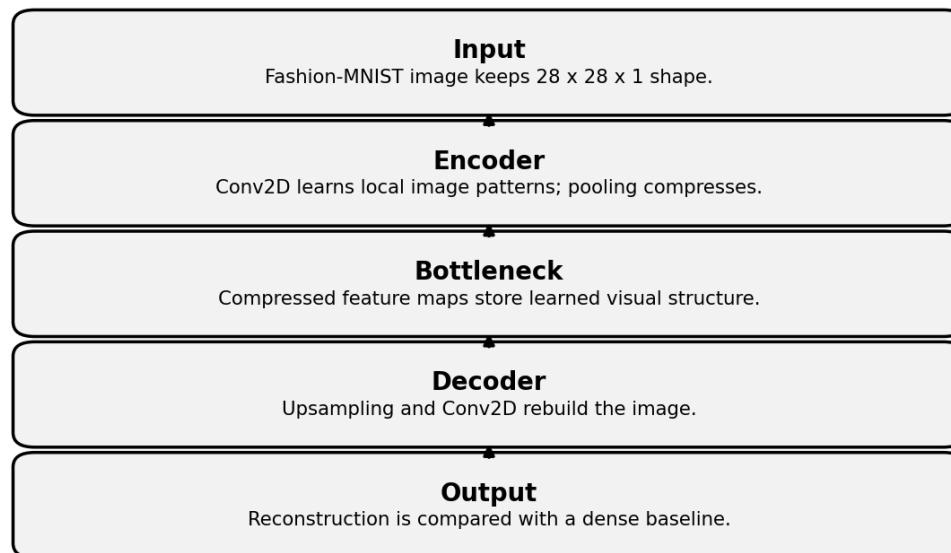
Dense Autoencoder flow diagram.

Source Variable	Method / Layer	Output Variable	Shape Change	Logic / Purpose
x_train, x_test	keras.datasets.mnist.load_data()	raw MNIST arrays	(60000,28,28), (10000,28,28)	Load grayscale digit images.
x_train	astype(float32) / 255.0	x_train_norm	(60000,28,28)	Normalize pixels to 0-1 for stable training.

Source Variable	Method / Layer	Output Variable	Shape Change	Logic / Purpose
x_train_norm	reshape(..., 784)	x_train_flat	(60000,784)	Flatten images for Dense layers.
input_image	Dense(128, relu)	encoder_dense_128 output	(batch,784)->(batch,128)	Learn hidden image features.
encoder_dense_128 output	Dense(32, relu)	bottleneck_32	(batch,128)->(batch,32)	Compress image into small representation.
bottleneck_32	Dense(128, relu)	decoder_dense_128 output	(batch,32)->(batch,128)	Start rebuilding from compressed code.
decoder_dense_128 output	Dense(784, sigmoid)	reconstructed_output	(batch,128)->(batch,784)	Reconstruct pixel values between 0 and 1.
x_test_flat	autoencoder.predict()	reconstructed_flat	(10000,784)	Generate reconstructions for unseen test images.
x_test_norm + reconstructed_images	np.mean(square difference)	reconstruction_errors	(10000,)	Measure per-image reconstruction error.

2. Convolutional Autoencoder on Fashion-MNIST

Convolutional Autoencoder Flow



Convolutional Autoencoder flow diagram.

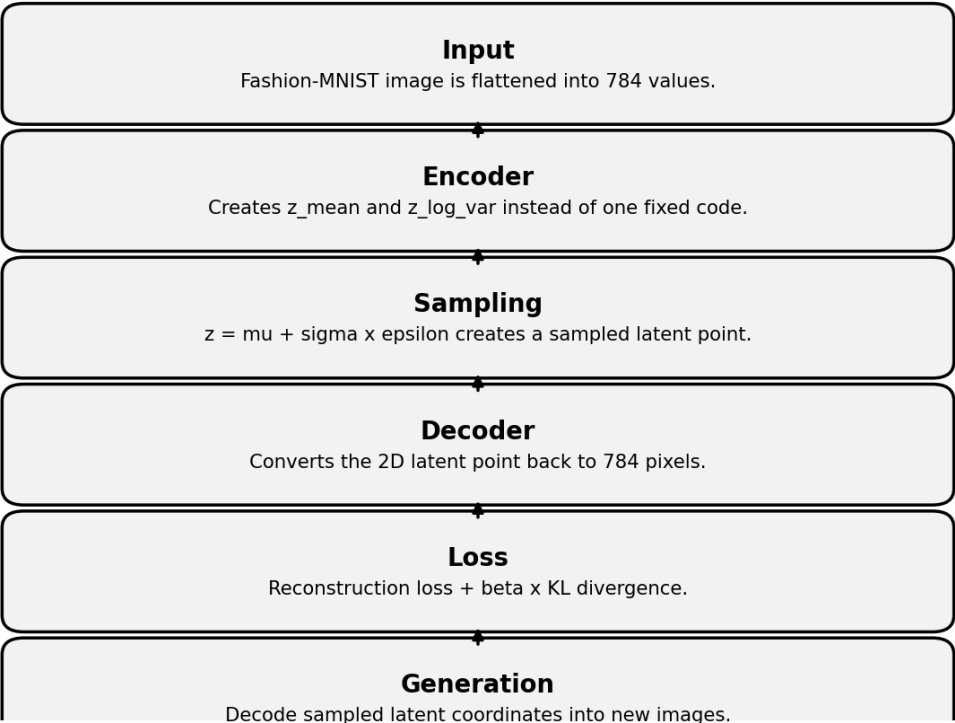
Source Variable	Method / Layer	Output Variable	Shape Change	Logic / Purpose
x_train, x_test	keras.datasets.fashion_mnist.load_data()	raw Fashion-MNIST arrays	(60000,28,28), (10000,28,28)	Load clothing images.
x_train	astype(float32) / 255.0	x_train_norm	(60000,28,28)	Normalize pixels to 0-1.
x_train_norm	np.expand_dims(axis=-1)	x_train_conv	(60000,28,28,1)	Add grayscale channel for Conv2D.
input_image	Conv2D(32, 3x3)	encoder_conv_32 output	(batch,28,28,1)->(batch,28,28,32)	Learn local visual patterns.
encoder_conv_32 output	MaxPooling2D(2x2)	encoder_pool_1 output	(batch,28,28,32)->(batch,14,14,32)	Compress spatial size.
encoder_pool_1 output	Conv2D(16, 3x3)	encoder_conv_16 output	(batch,14,14,32)->(batch,14,14,16)	Learn deeper compressed features.
encoder_conv_16 output	MaxPooling2D(2x2)	bottleneck_7x7x16	(batch,14,14,16)->(batch,7,7,16)	Compressed image feature map.
bottleneck_7x7x16	Conv2D + UpSampling	reconstructed_output	(batch,7,7,16)->(batch,28,28,1)	Rebuild image structure.
bottleneck_features	reshape((n,-1))	bottleneck_features_flat	(1000,7,7,16)->(1000,784)	Prepare encoder features for PCA/t-SNE/UMAP.

3. Dense Baseline Used in Intermediate Notebook

Source Variable	Method / Layer	Output Variable	Shape Change	Logic / Purpose
x_train_conv	reshape(..., 784)	x_train_flat_fashion	(60000,28,28,1)->(60000,784)	Flatten Fashion-MNIST for Dense baseline.
dense_fashion_input	Dense(128) -> Dense(32)	dense_bottleneck_32	(batch,784)->(batch,32)	Create comparable Dense bottleneck.
dense_bottleneck_32	Dense(128) -> Dense(784)	dense_reconstructed_output	(batch,32)->(batch,784)	Reconstruct Fashion-MNIST without 2D structure.
same 100 test images	predict + MSE	comparison_df	two model rows	Compare Dense vs Conv reconstruction errors fairly.

4. Variational Autoencoder on Fashion-MNIST

Variational Autoencoder Flow



VAE flow diagram.

Source Variable	Method / Layer	Output Variable	Shape Change	Logic / Purpose
x_train	astype(float32)/255 + reshape	x_train_flat	(60000,28,28)->(60000,784)	Prepare flattened images for Dense VAE.
encoder_inputs	Dense(128, relu)	x	(batch,784)->(batch,128)	Learn hidden image features.
x	Dense(2)	z_mean	(batch,128)->(batch,2)	Centre of latent distribution.
x	Dense(2)	z_log_var	(batch,128)->(batch,2)	Spread/uncertainty of latent distribution.
z_mean + z_log_var	Sampling layer	z	(batch,2)	Apply reparameterisation trick.
z	decoder Dense layers	reconstruction	(batch,2)->(batch,784)	Rebuild Fashion-MNIST image.
x + reconstruction	binary_crossentropy * input_dim	reconstruction_loss	scalar	Measure image reconstruction error.
z_mean + z_log_var	KL formula	kl_loss	scalar	Regularise latent space.

Source Variable	Method / Layer	Output Variable	Shape Change	Logic / Purpose
reconstruction_loss + beta * kl_loss	weighted sum	total_loss	scalar	Final VAE training objective.

5. Custom VAE Class Method Flow

Method	Called By	Important Parameters	What It Does	Why It Matters
__init__()	Automatically when VAE(...) is created	encoder, decoder, beta, self	Stores encoder, decoder, beta, and loss trackers inside the VAE object.	Sets up the full model and training state.
metrics property	Keras automatically	self	Tells Keras which metrics to reset each epoch.	Keeps epoch logging clean.
compute_vae_loss(x)	Called manually inside train_step/test_step	x = current image batch	Runs encoder, decoder, reconstruction loss, KL loss, and total loss.	Central loss logic reused for training and validation.
train_step(data)	Keras automatically during model.fit()	data = one training batch	Computes loss, gradients, updates weights, updates logs.	Defines one VAE training step.
test_step(data)	Keras automatically during validation	data = one validation batch	Computes validation losses without updating weights.	Allows correct validation loss logging.

6. Key Result Variables

Variable	Created In	Meaning	Final/Observed Value
reconstruction_errors	Beginner notebook	MNIST per-image MSE	Average 0.008549
conv_reconstruction_errors	Intermediate notebook	Conv AE per-image MSE	Average 0.004471
comparison_df	Intermediate notebook	Dense vs Conv reconstruction comparison	Conv reduced error by about 82.3%
history_beta_1_df	Expert notebook	beta = 1 VAE training history	Final total loss 288.969452
history_beta_4_df	Expert notebook	beta = 4 VAE training history	Final total loss 312.896362
reconstruction_errors_beta_1	Expert notebook	beta = 1 VAE reconstruction MSE	Average 0.039027
reconstruction_errors_beta_4	Expert notebook	beta = 4 VAE reconstruction MSE	Average 0.044348
generated_grid	Expert notebook	Latent-space generated image grid	15 x 15 generated Fashion-MNIST-like images

Source note: This variable flow document is based on the three uploaded notebooks completed for Learning Day 6.